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Control Systems — Master Formula Sheet & Stability Guide

Mason's Rule, Second-Order Time Specs, Routh-Hurwitz, Root Locus, Bode & Nyquist Criteria

GATE Section: Section 6 • **Marks:** 8 - 10 Marks • **Core Focus:** System Modeling, Frequency Response, State Space

TIME-DOMAIN SPECS & COMPENSATOR RULES
- **Maximum Peak Overshoot:** Mp = exp(-pi * zeta / sqrt(1 - zeta^2)) * 100% (Depends ONLY on damping ratio zeta!).
- **Settling Time:** ts = 4 / (zeta * omega_n) for 2% tolerance band; ts = 3 / (zeta * omega_n) for 5% band.
- **Lead Compensator:** Adds positive phase, increases bandwidth, improves transient speed. Dominant zero closer to origin.

SECTION: MASON'S GAIN FORMULA & SECOND-ORDER TIME RESPONSE

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Mason's Gain Formula	**T = (1 / Delta) * sum_{k} [P_k * Delta_k]**	P_k = gain of k-th forward path Delta = 1 - sum(L1) + sum(L2) - sum(L3)...	Delta_k = value of Delta for path that does not touch k-th forward path.
Characteristic Equation	**s^2 + 2*zeta*omega_n*s + omega_n^2 = 0**	Poles: s = -zeta*omega_n +- j*omega_d Damped frequency: omega_d = omega_n*sqrt(1 - zeta^2)	Angle of pole: theta = arccos(zeta). Real part determines settling speed; imaginary part determines oscillation frequency.
Peak Time & Rise Time	**t_p = pi / omega_d = pi / (omega_n * sqrt(1 - zeta^2))** **t_r = (pi - beta) / omega_d**	beta = arctan(sqrt(1 - zeta^2) / zeta) in radians	Peak time is the time taken to reach the first maximum peak overshoot.
Static Error Constants	**K_p = lim_{s->0} G(s)H(s) (Position)** **K_v = lim_{s->0} s*G(s)H(s) (Velocity)** **K_a = lim_{s->0} s^2*G(s)H(s) (Acceleration)**	Type 0: Step error ess = 1/(1+Kp), Ramp error = inf Type 1: Step error = 0, Ramp error = 1/Kv Type 2: Step = 0, Ramp = 0, Parabolic = 1/Ka	Steady state error ess = lim_{s->0} s * R(s) / (1 + G(s)H(s)).

SECTION: STABILITY & ROOT LOCUS RULES

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Routh-Hurwitz Array	**First column has no sign changes -> Stable** **Row of all zeros -> Symmetric poles about origin**	Auxiliary equation A(s) formed from previous row Take dA/ds to replace zero row	Number of sign changes in first column = Number of roots in Right-Half s-plane (unstable poles).
Root Locus Angles	**Asymptote angles: phi_A = (2q + 1)*180 / (P - Z)** **Centroid: sigma_A = (sum(Re(poles)) - sum(Re(zeros))) / (P - Z)**	P = number of poles, Z = number of zeros q = 0, 1, ..., (P - Z - 1)	Root locus branches start at open-loop poles (K = 0) and terminate at open-loop zeros or infinity (K = inf).
Breakaway Points	**dK / ds = 0**	Valid breakaway point must lie on root locus segment!	Breakaway angle = 180 / n where n is number of branches arriving at point.

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Angle of Departure	**phi_d = 180 - sum(angles from other poles) + sum(angles from zeros)**	Computed for complex conjugate open-loop poles	Angle of arrival at complex zeros: phi_a = 180 - sum(angles from other zeros) + sum(angles from poles).

SECTION: BODE, NYQUIST & STATE-SPACE FORMULAS

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Gain & Phase Margins	**GM = -20 * log10 |G(j*omega_pc)| in dB** **PM = 180 + angle(G(j*omega_gc)) in degrees**	omega_pc: angle(G(j*omega_pc)) = -180 deg omega_gc: |G(j*omega_gc)| = 1	For stable minimum phase system: omega_gc < omega_pc (GM > 0 dB, PM > 0 deg).
Nyquist Criterion	**N = P - Z** **Z = P - N = 0 (for closed-loop stability)**	N = counter-clockwise encirclements of -1+j0 P = open-loop poles in RHP, Z = closed-loop poles in RHP	If open-loop system is stable (P = 0), closed-loop is stable iff N = 0 (no encirclements).
State Transition Matrix	**phi(t) = exp(A*t) = L^-1{(sI - A)^-1}**	phi(0) = I, phi(t1+t2) = phi(t1)*phi(t2) [phi(t)]^-1 = phi(-t)	Solution: x(t) = phi(t)*x(0) + integral_0^t phi(t - tau)*B*u(tau) dtau.
Controllability & Observability	**Q_c = [B | A*B | A^2*B | ... | A^(n-1)*B]** **Q_o = [C^T | A^T*C^T | ... | (A^T)^(n-1)*C^T]**	System controllable iff rank(Q_c) = n System observable iff rank(Q_o) = n	Transfer function: T(s) = C * (sI - A)^-1 * B + D.